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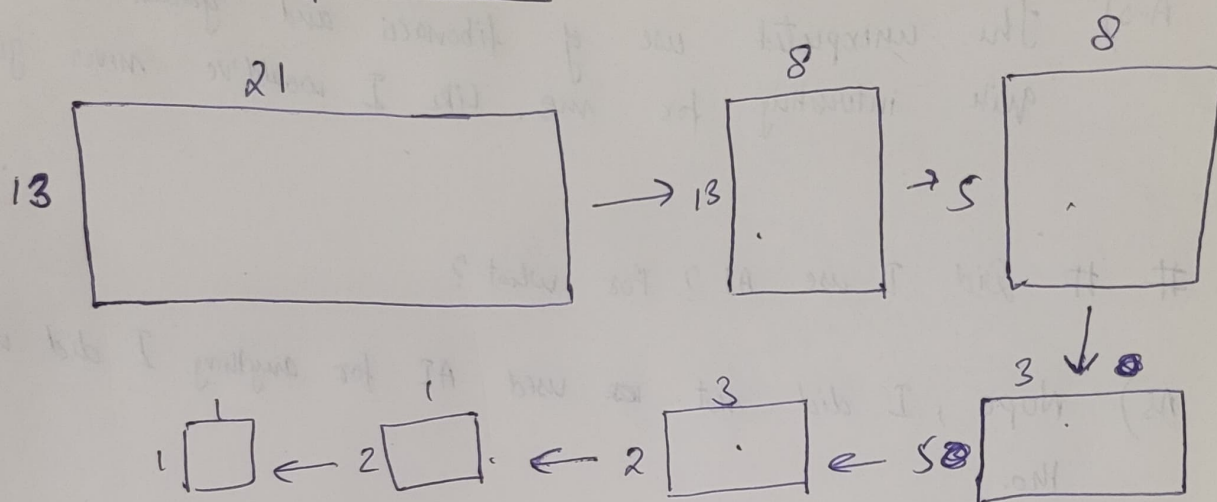
Enigma number: 122344

My guess: 13×8 would have the highest steps

My N: 33

(a, b)	No. of steps	GCD
$(5, 7)$	3	1
$(8, 17)$	3	1
$(5, 8)$	4	1
$(6, 15)$	2	3
$(13, 21)$	6	1

• Slowest pair and its steps -



Why is it slow -

- All the fibonacci pairs ~~not~~ would have the most steps and amongst them, (13, 21) has the highest number of ~~step~~ steps under 33, this is because ~~the~~ the max that can be cut is 1.
- The findings was found using a simple python program which is also attached.
- Explanation - fibonacci sequence would always have the ratio of ≈ 2 (two continuous pairs), therefore the max that can be cut is just one time. and then after cutting, the next dimension formed is also fibonacci seq.

$$2 \times 1 \rightarrow 1$$

$$2 \times 3 \rightarrow \text{cut one } 2 \times 2 \text{ sq, leaves } 2 \times 1$$

$$5 \times 3 \rightarrow \text{cut one } 3 \times 3, \text{ leaves } 2 \times 3$$

$$8 \times 5 \rightarrow \text{cut one } 5 \times 5, \text{ leaves } 3 \times 3$$

recurrence
~~forming~~
forming

What surprised me?

Ans) The unexpected use of fibonacci and golden ratio was quite interesting for me. Like I would've never guessed it.

Did I use AI? For what?

Ans) Nope, I did not ~~use~~ used AI for anything. I did use python tho.